



6.2.7 Wrap-Up: Error Analysis

1. Simeon solved a compound inequality. His work is shown.

$$18 < -3y + 72 < \frac{1}{3}(168y - 92)$$

$$\begin{array}{lcl} & \swarrow & \searrow \\ 18 < -3y + 72 & & 18 < \frac{1}{3}(168y - 92) \\ -54 < -3y & & 18 < 56y - 46 \\ 18 > y & & 54 < 56y \\ & & \frac{54}{56} < y \\ & & \frac{27}{28} < y \end{array}$$

Simeon stated the solution set is $\frac{27}{28} < y < 18$.

Explain the mistake(s) Simeon made.

Unit 6, Lesson 3: Writing and Solving Inequalities



Benchmark

MA.912.AR.2.6 Given a mathematical or real-world context, write and solve one-variable linear inequalities, including compound inequalities. Represent solutions algebraically or graphically.



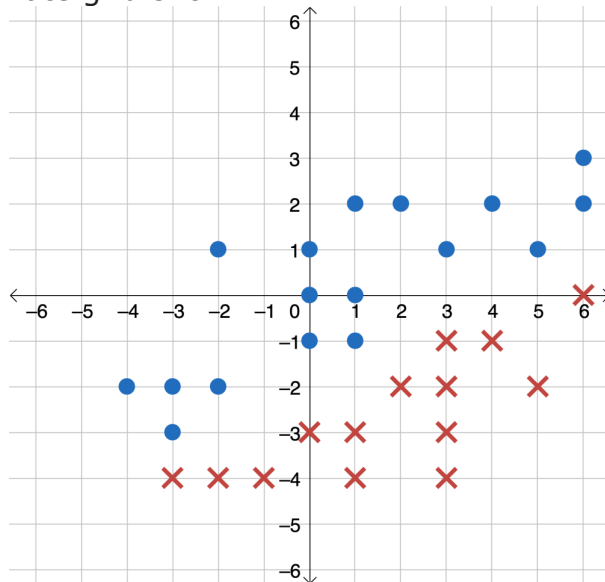
Learning Targets

- ☐ I can write and solve a two-step, one-variable inequality.
- ☐ I can write and solve a multi-step, one-variable inequality.
- ☐ I can write and solve a compound one-variable inequality.
- ☐ I can interpret solutions to problems as viable or non-viable options.



6.3.1 Warm-Up: Notice and Wonder

1. Draw a line that separates the **circles** and the **Xs** on the coordinate grid shown.



- a. Find any **X** whose x -coordinate is greater than the x -coordinate of a circle. Write their ordered pairs.

X	●

- b. What do you notice about the y -coordinates?

- c. Find any **X** whose x -coordinate is the same as the x -coordinate of a circle. Write their ordered pairs.

X	●

- d. What do you notice about the y -coordinates?



6.3.2 Collaboration: Inequalities in Real-World Contexts

Work with your partner to complete the following.

1. Three inequalities are given.

$$5h + 4.5 \leq 20$$

$$4.5 \leq 5k \leq 20$$

$$5m > 20 + 4.5m$$

Match each of the inequalities to one of the situations below. Then, solve the inequality and explain how it matches the situation.

Situation A

Markus went to his local coffee shop to get a latte and some coffee beans. The latte cost \$4.50 and beans cost \$5 per pound. He has a \$20 bill and wants to know how many pounds of beans he can buy.

Matching Inequality and Solution

How does the inequality match the situation?

Situation B	
At Rolling Lanes Bowling Alley, the cost is \$5 per game for non-members and \$4.50 per game for members. There is a \$20 annual fee to be a member. Luna wants to know how many games she needs to play in a year in order for the membership to be a good deal.	
Matching Inequality and Solution	How does the inequality match the situation?
Situation C	
Ihab is training for a marathon. He runs at a pace of 5 miles per hour (mph). In the past week, the distances of his runs have ranged from 4.5 miles to 20 miles. He writes an inequality that can be used to find the range of time he's spent running in the past week.	
Matching Inequality and Solution	How does the inequality match the situation?



6.3.3 Exploration: Is There Enough Time?

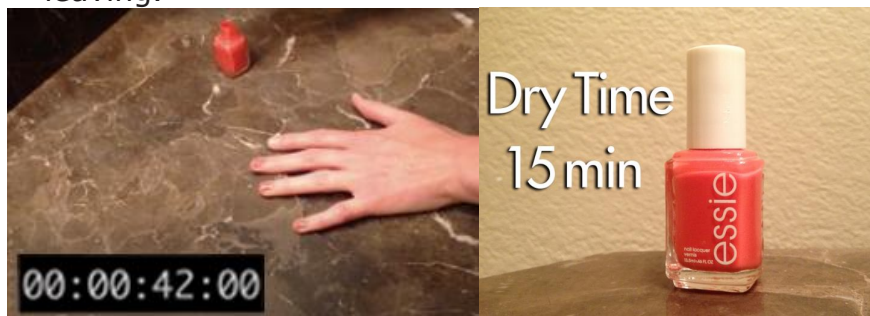
1. Candace is getting ready to leave for the movies. She decides at the last minute to try to paint her nails before she leaves.



It’s 6:40 p.m., and Candace needs to leave at 7:00 p.m. What information do you need to determine if she will have enough time to paint her nails?



2. The images shown give additional information about the situation. The first image shows it takes her 42 seconds to paint the first three nails. The second image gives the amount of time she needs to let her nails dry before leaving.



- How long does it take her to paint each nail, on average?
- Explain whether or not you think Candace will have enough time to paint her nails.



6.3.4 Guided Instruction: Writing Inequalities to Represent Real-World Contexts

- The nail polish scenario provides an example of how inequalities can be used to solve problems in the real world.
 - Let t be the time it takes to paint one nail, in minutes. Write an expression in terms of t that represents the total amount of time it takes for the nail polish process to be complete, including 15 minutes for drying.
 - In the Exploration, Candace had a maximum of 20 minutes. Using your expression from Part A, write an inequality that represents this situation.
 - Solve the inequality.
 - Explain how the maximum value in the solution set relates to the nail polish situation.
 - Graph the solution set.



f. The value -0.5 is in the solution set of the inequality. Discuss with your partner why this is not a viable solution in the real-world context. Write your explanation.

g. Explain whether or not the value 0 is a viable solution in this context.

h. Select any statements that apply for values to be viable solutions in this context.

- ☐ Values must be non-negative.
- ☐ Values must be greater than 0 .
- ☐ Values must be integers.
- ☐ Values cannot be more than ____ decimal places.
- ☐ Values cannot be greater than 0.5 .

i. Complete the compound inequality that represents the full set of possible solutions in the real-world context.

$$\underline{\hspace{1cm}} < t \leq \underline{\hspace{1cm}}$$



6.3.5 Activity: Gallery Walk

Work with your partner or small group to complete the following for each real-world context.

- Write an inequality to represent the situation, including defining the variable used.
- Solve the inequality and graph the solution.
- Identify and explain any non-viable solutions.

Context A – Vacation Savings	
Inequality	
Solution	
Graph	
Non-Viable Solutions in the Set	Check any that apply. ☐ Solutions must be non-negative ☐ Solutions must be integers Explain.

Context B – Non-Operating Temperatures	
Inequality	
Solution	
Graph	
Non-Viable Solutions in the Set	Check any that apply. ☐ Solutions must be non-negative ☐ Solutions must be integers Explain.

Context C – Gym Memberships	
Inequality	
Solution	
Graph	
Non-Viable Solutions in the Set	Check any that apply. ☐ Solutions must be non-negative ☐ Solutions must be integers Explain.

Context D – Planting Citrus Trees

Inequality	
Solution	
Graph	
Non-Viable Solutions in the Set	Check any that apply. ☐ Solutions must be non-negative ☐ Solutions must be integers Explain.



6.3.6 Wrap-Up: Reflection

Which mindset did I demonstrate?	Choose the best description.
Did I use the partner discussions as learning opportunities?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I work on problems that challenged me?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I use strategies to “un-stick” myself when I found the problems difficult?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I re-read the problem to ensure understanding of the context?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I put as much effort as I possibly could into the problems?	☐ Never ☐ Sometimes ☐ Usually ☐ Always